

Application of Definite Integral:

Definite integral can be used to find area under the curve, volume of solids and many useful things. Here it is used to find area under curve.

Area above x-axis is $\int_a^b y dx$

Area below x-axis is $\int_a^b -y dx$

EXERCISE 3.7

Q.1 Find the area between the x-axis and the curve $y = x^2 + 1$ from $x = 1$ to $x = 2$.

Solution:

$$y = x^2 + 1, \quad a = 1, \quad b = 2,$$

Since area lies above x-axis

$$\begin{aligned} \text{Area} &= \int_a^b y dx \\ &= \int_1^2 (x^2 + 1) dx = \int_1^2 x^2 dx + \int_1^2 1 dx \\ &= \left[\frac{x^3}{3} \right]_1^2 + [x]_1^2 \\ &= \frac{1}{3} (2^3 - 1^3) + (2 - 1) \\ &= \frac{1}{3} (8 - 1) + 1 \\ &= \frac{3}{7} + 1 = \frac{10}{3} \text{ Square units} \end{aligned}$$

Q.2 Find the area above the x-axis and under the curve $y = 5 - x^2$ from $x = -1$ to $x = 2$.

Solution:

$$y = 5 - x^2, \quad a = -1, \quad b = 2$$

Since Area lies above x-axis

$$\begin{aligned} \text{Area} &= \int_a^b f(x) dx \\ &= \int_{-1}^2 (5 - x^2) dx \end{aligned}$$

$$\begin{aligned} &= \int_{-1}^2 5 dx - \int_{-1}^2 x^2 dx \\ &= [5x]_{-1}^2 - \left[\frac{x^3}{3} \right]_{-1}^2 \\ &= 5(2 - (-1)) - \frac{1}{3} (2^3 - (-1)^3) \\ &= 5(3) - \frac{1}{3} (8 + 1) \\ &= 15 - 3 \\ &= 12 \text{ Square units} \end{aligned}$$

Q.3 Find the area below the curve $y = 3\sqrt{x}$ and above the x-axis between $x = 1$ and $x = 4$

Solution:

$$\text{Let } y = 3\sqrt{x}, \quad a = 1, \quad b = 4$$

Since area lies above x-axis

$$\begin{aligned} \text{Area} &= \int_a^b y dx \\ &= \int_1^4 3\sqrt{x} dx \\ &= 3 \int_1^4 x^{1/2} dx = 3 \left[\frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right]_1^4 \\ &= 3 \cdot \left[\frac{\frac{3}{2}}{\frac{3}{2}} \right]_1^4 = 3 \cdot \frac{2}{3} \left[x^{\frac{3}{2}} \right]_1^4 \end{aligned}$$

$$= 2 \left((4)^{\frac{3}{2}} - (1)^{\frac{3}{2}} \right)$$

$$= 2(8-1) = 14 \text{ Square units}$$

Q.4 Find the area bounded by cos

function from $x = -\frac{\pi}{2}$ to $x = \frac{\pi}{2}$

Solution:

Let $y = \cos x$, $a = -\frac{\pi}{2}$, $b = \frac{\pi}{2}$

Since area lies above x-axis

$$\text{Area} = \int_a^b y \, dx$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \, dx$$

$$= \left[\sin x \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= \sin \frac{\pi}{2} - \sin \left(-\frac{\pi}{2} \right)$$

$$= 1 - (-1) = 2 \text{ Square units}$$

Q.5 Find the area between the x-axis and the curve $y = 4x - x^2$

Solution:

Let $y = 4x - x^2$

To find out the limits, let $y = 0$

$$4x - x^2 = 0$$

$$x(4 - x) = 0$$

$$x = 0, x = 4$$

Since area lies above x-axis

$$\text{Area} = \int_a^b y \, dx$$

$$= \int_0^4 (4x - x^2) \, dx$$

$$= \int_0^4 4x \, dx - \int_0^4 x^2 \, dx$$

$$= 4 \int_0^4 x \, dx - \int_0^4 x^2 \, dx$$

$$= 4 \left[\frac{x^2}{2} \right]_0^4 - \left[\frac{x^3}{3} \right]_0^4$$

$$= 2 \left[x^2 \right]_0^4 - \frac{1}{3} \left[x^3 \right]_0^4$$

$$= 2(4^2 - 0^2) - \frac{1}{3}(4^3 - 0^3)$$

$$= 2(16) - \frac{1}{3}(64)$$

$$= 32 - \frac{64}{3} = \frac{96 - 64}{3}$$

$$= \frac{32}{3} \text{ Square units}$$

Q.6 Determine the area bounded by the parabola $y = x^2 + 2x - 3$ and the x-axis.

Solution:

Let $y = x^2 + 2x - 3$

To find out the limits, let $y = 0$

$$x^2 + 2x - 3 = 0$$

$$x^2 + 3x - x - 3 = 0$$

$$x(x+3) - 1(x+3) = 0$$

$$(x-1)(x+3) = 0$$

$$x = 1, x = -3$$

Since area lies below x-axis

$$\text{Area} = - \int_a^b y \, dx$$

$$= - \int_{-3}^1 (x^2 + 2x - 3) \, dx$$

$$= - \left[\frac{x^3}{3} + x^2 - 3x \right]_{-3}^1$$

$$= - \left(\frac{1}{3} + 1 - 3 \right) + \left(-\frac{27}{3} + 9 + 9 \right)$$

$$= - \left(\frac{1}{3} - 2 \right) + 9$$

$$= \frac{5}{3} + 9$$

$$= \frac{5 + 27}{3}$$

$$= \frac{32}{3} \text{ Square units}$$

Q.7 Find the area bounded by the curve

$y = x^3 + 1$, the x -axis and the line
 $x = 2$.

Solution:

$$y = x^3 + 1, x = 2$$

To find out the limits, let $y = 0$

$$x^3 + 1 = 0 \Rightarrow (x+1)(x^2 - x + 1) = 0$$

Either $x+1=0 \Rightarrow x=-1$

Or $x^2 - x + 1 = 0$ (neglected
because it gives complex roots)

Since area lies above x -axis

$$\begin{aligned} \text{Area} &= \int_a^b f(x) dx \\ &= \int_{-1}^2 (x^3 + 1) dx = \int_{-1}^2 x^3 dx + \int_{-1}^2 1 dx \\ &= \left[\frac{x^4}{4} \right]_{-1}^2 + [x]_{-1}^2 \\ &= \frac{1}{4} (2^4 - (-1)^4) + (2 - (-1)) \\ &= \frac{1}{4} (16 - 1) + 3 \\ &= \frac{15}{4} + 3 \\ &= \frac{15 + 12}{4} \\ &= \frac{27}{4} \text{ Square units} \end{aligned}$$

Q.8 Find the area bounded by the curve

$y = x^3 - 4x$ and the x -axis.

Solution:

$$y = x^3 - 4x$$

To find out the limits, let $y = 0$

$$x^3 - 4x = 0$$

$$x(x^2 - 4) = 0$$

$$x = 0, x = \pm 2$$

Since area lies above x -axis for
 $-2 < x < 0$

$$A_1 = \int_a^b y dx$$

$$\begin{aligned} &= \int_{-2}^0 (x^3 - 4x) dx \\ &= \left[\frac{x^4}{4} \right]_{-2}^0 - \left[\frac{4x^2}{2} \right]_{-2}^0 \\ &= \frac{1}{4} (0 - (-2)^4) - 2(0 - (-2)^2) \\ A_1 &= \frac{1}{4} (-16) - 2(-4) = -4 + 8 = 4 \end{aligned}$$

Since area lies below x -axis for
 $0 < x < 2$

$$\begin{aligned} A_2 &= - \int_a^b y dx \\ &= - \int_0^2 (x^3 - 4x) dx \\ &= - \int_0^2 x^3 dx + 4 \int_0^2 x dx \\ &= - \left[\frac{x^4}{4} \right]_0^2 + 4 \left[\frac{x^2}{2} \right]_0^2 \\ &= \frac{-1}{4} (16 - 0) + 2(4 - 0) \end{aligned}$$

$$A_2 = -4 + 8 = 4$$

$$\text{Total area} = A_1 + A_2$$

$$A = 4 + 4 = 8 \text{ Square units}$$

Q.9 Find the area between the curve

$y = x(x-1)(x+1)$ and the x -axis.

Solution:

$$\begin{aligned} y &= x(x-1)(x+1) \\ &= x^3 - x \end{aligned}$$

To find out the limits, let $y = 0$

$$x(x-1)(x+1) = 0$$

$$x = 0, x - 1 = 0, x + 1 = 0$$

$$x = 0, x = 1, x = -1$$

Since area lies above x -axis for
 $-1 < x < 0$

$$A_1 = \int_a^b y dx$$

$$= \int_{-1}^0 (x^3 - x) dx$$

$$= \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0$$

$$= 0 - 0 - \left(\frac{1}{4} - \frac{1}{2} \right)$$

$$A_1 = -\frac{1}{4} + \frac{1}{2} = \frac{-1+2}{4} = \frac{1}{4}$$

Since area lies below x -axis for $0 < x < 1$

$$A_2 = -\int_a^b y dx$$

$$= -\int_0^1 (x^3 - x) dx$$

$$= \int_0^1 (-x^3 + x) dx$$

$$= \left[-\frac{x^4}{4} + \frac{x^2}{2} \right]_0^1$$

$$= -\frac{1}{4} + \frac{1}{2} - (0+0)$$

$$A_2 = \frac{-1+2}{4} = \frac{1}{4}$$

$$\text{Total area} = A_1 + A_2$$

$$A = \frac{1}{4} + \frac{1}{4} = \frac{1+1}{4} = \frac{2}{4} = \frac{1}{2} \text{ Square units}$$

Q.10 Find the area above the x -axis, bounded by the curve $y^2 = 3-x$ from $x = -1$ to $x = 2$

Solution:

$$y^2 = 3-x$$

$$\Rightarrow y = \pm \sqrt{3-x}$$

$$y = \sqrt{3-x}, a = -1, b = 2$$

Since area lies above x -axis

$$\text{Area} = \int_a^b y dx$$

$$= \int_{-1}^2 \sqrt{3-x} dx$$

$$= -\int_{-1}^2 (3-x)^{\frac{1}{2}} (-1) dx$$

$$\begin{aligned} &= -\left[\left(\frac{3-x}{\frac{3}{2}} \right)^{\frac{3}{2}} \right]_{-1}^2 = -\frac{2}{3} \left[(3-x)^{\frac{3}{2}} \right]_{-1}^2 \\ &= -\frac{2}{3} \left[(3-2)^{\frac{3}{2}} - (3+1)^{\frac{3}{2}} \right] \\ &= -\frac{2}{3} \left[(1)^{\frac{3}{2}} - (4)^{\frac{3}{2}} \right] \\ &= -\frac{2}{3} (1-8) \\ &= -\frac{2}{3} (-7) \\ &= \frac{14}{3} \text{ Square units} \end{aligned}$$

Q.11 Find the area between the x -axis and the curve $y = \cos \frac{x}{2}$ from $x = -\pi$ to π .

Solution:

$$y = \cos \frac{x}{2}, a = -\pi, b = \pi$$

Since area lies above x -axis

$$\text{Area} = \int_a^b y dx$$

$$= \int_{-\pi}^{\pi} \cos \frac{x}{2} dx$$

$$= \left[\frac{\sin \frac{x}{2}}{\frac{1}{2}} \right]_{-\pi}^{\pi}$$

$$= 2 \left[\sin \left(\frac{\pi}{2} \right) - \sin \left(-\frac{\pi}{2} \right) \right]$$

$$= 2[1 - (-1)] = 2(2)$$

$$= 4 \text{ Square units}$$

Q.12 Find the area between the x -axis and the curve $y = \sin 2x$ from $x = 0$ to

$$x = \frac{\pi}{3}$$

Solution:

$$y = \sin 2x \quad a = 0, \quad b = \frac{\pi}{3}$$

Since area lies above x -axis

$$\begin{aligned}
 \text{Area} &= \int_a^b f(x) dx \\
 &= \int_0^{\frac{\pi}{3}} \sin 2x dx \\
 &= -\left[\frac{\cos 2x}{2} \right]_0^{\frac{\pi}{3}} \\
 &= -\frac{1}{2} \left[\cos \frac{2\pi}{3} - \cos 0 \right] \\
 &= -\frac{1}{2} \left[-\frac{1}{2} - 1 \right] \\
 &= -\frac{1}{2} \left(-\frac{3}{2} \right) \\
 &= \frac{3}{4} \text{ Square units}
 \end{aligned}$$

Q.13 Find the area between the x-axis and the curve $y = \sqrt{2ax - x^2}$ when $a > 0$

Solution:

$$y = \sqrt{2ax - x^2}$$

To find out the limits, let $y = 0$

$$2ax - x^2 = 0$$

$$x(2a - x) = 0$$

$$x = 0, 2a - x = 0$$

$$x = 0, x = 2a$$

Since area lies above x-axis

$$\begin{aligned}
 \text{Area} &= \int_a^b y dx \\
 &= \int_0^{2a} \sqrt{2ax - x^2} dx \\
 &= \int_0^{2a} \sqrt{a^2 + 2ax - x^2 - a^2} dx \\
 &= \int_0^{2a} \sqrt{a^2 - (x^2 - 2ax + a^2)} dx \\
 &= \int_0^{2a} \sqrt{a^2 - (x - a)^2} dx
 \end{aligned}$$

$$\text{Put } x - a = a \sin \theta \Rightarrow dx = a \cos \theta d\theta$$

$$\text{when } x \rightarrow 0, \quad \theta \rightarrow \frac{-\pi}{2}$$

$$\text{when } x \rightarrow 2a, \quad \theta \rightarrow \frac{\pi}{2}$$

$$\begin{aligned}
 &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{a^2 - a^2 \sin^2 \theta} a \cos \theta d\theta \\
 &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{a^2 (1 - \sin^2 \theta)} a \cos \theta d\theta \\
 &= a^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{1 - \sin^2 \theta} \cos \theta d\theta \\
 &= a^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 \theta d\theta \\
 &= a^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1 + \cos 2\theta}{2} \right) d\theta \\
 &= \frac{a^2}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 d\theta + \frac{a^2}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos 2\theta d\theta \\
 &= \frac{a^2}{2} \left[\theta \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \frac{a^2}{2} \left[\frac{\sin 2\theta}{2} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\
 &= \frac{a^2}{2} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) + \frac{a^2}{4} (\sin \pi - \sin(-\pi)) \\
 &= \frac{a^2}{2} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) + \frac{a^2}{4} (0 - 0) \\
 &= \frac{a^2}{2} (\pi) \\
 &= \frac{1}{2} \pi a^2 \text{ Square units}
 \end{aligned}$$